HW2

Dongwook Hwang

Q1.

	Models	AIC	BIC
1	POW1	475.213	475.292
2	POW2	477.213	477.371
3	EXP1	524.452	524.532
4	EXP2	489.176	489.335
5	EXPOW	479.213	479.451
6	HYP1	482.875	482.954
7	HYP2	481.475	481.634

The AIC and BIC values were computed for the seven retention models from HW1, using the maximun likelihood estimates obtained previously. Among the seven models, POW1 showed the lowest AIC (475.213) and BIC (475.292), indicating that it provides the best balance between model fit and complexity. POW2 (AIC = 477.213) and EXPOW (AIC = 479.213) followed closely, with differences of approximately 2 and 4 points from POW1, respectively. This pattern reflects how AIC penalizes additional parameters: although POW1, POW2, and EXPOW all achieved the same R^2 (0.980) in HW1, the more complex models receive higher penalties without improving fit. EXP1 showed by far the worst performance (AIC = 524.452), consistent with the HW1 finding that purely exponential decay does not capture the retention data well. Overall, the AIC/BIC analysis confirms that POW1 is the most appropriate model.

Q2.

```
[1] "--- ra_prospect ---"
            rho    lambda      tau
subj_2 1.0198788 0.7827932 1.007139
subj_3 0.6926625 2.4746740 4.742068
subj_4 0.7944019 1.0765146 1.035454
subj_6 0.8655623 0.9823003 3.129862
subj_7 0.9616153 1.4085401 2.346806
```

ρ: risk preference, λ = loss aversion, τ = inverse temperature

For subj_2 alone, the estimated parameters were $\rho = 1.020$, $\lambda = 0.783$, and $\tau = 1.007$. ρ near 1 indicates minimal risk aversion, while $\lambda < 1$ suggests that this subject was less sensitive to losses than to gains. The estimation was extended to all five subjects using a similar procedure. Across subjects, ρ ranged from

0.69 to 1.02, with most subjects showing slight risk aversion ($\rho < 1$). λ varied substantially (0.78 to 2.47), indicating individual differences in loss aversion, and τ ranged from 1.01 to 4.74, reflecting different degrees of choice consistency.

Q3.

```
[1] "--- ra_prospect ---"
             rho    lambda      tau
subj_2 1.0198788 0.7827932 1.007139
subj_3 0.6926625 2.4746740 4.742068
subj_4 0.7944019 1.0765146 1.035454
subj_6 0.8655623 0.9823003 3.129862
subj_7 0.9616153 1.4085401 2.346806
[1] "--- ra_prospect (per subject) ---"
  Subject     AIC     BIC
1  subj_2 143.358 152.183
2  subj_3  42.934  51.759
3  subj_4 168.037 176.862
4  subj_6  93.589 102.414
5  subj_7  88.390  97.215
[1] "--- Model Comparison (summed across subjects) ---"
        Model sum_AIC sum_BIC
1 ra_prospect 536.309 580.434
```

```
[1] "--- ra_noLA ---"
             rho       tau
subj_2 0.9908216 0.7479853
subj_3 0.0000000 2.2802050
subj_4 0.8076192 1.0807460
subj_6 0.8668039 3.0491938
subj_7 0.9710553 1.3537039
[1] "--- ra_noLA (per subject) ---"
  Subject     AIC     BIC
1  subj_2 152.963 158.846
2  subj_3  90.548  96.431
3  subj_4 166.802 172.685
4  subj_6  91.783  97.666
5  subj_7 120.354 126.237
[1] "--- Model Comparison (summed across subjects) ---"
    Model sum_AIC sum_BIC
1 ra_noLA 622.449 651.866
```

```
[1] "--- ra_noRA ---"
           lambda       tau
subj_2  0.7881032 1.0774286
subj_3 10.0000000 0.2110716
subj_4  1.0990261 0.4624106
subj_6  1.0013289 1.6802363
subj_7  1.4344641 1.9983549
[1] "--- ra_noRA (per subject) ---"
  Subject     AIC     BIC
1  subj_2 141.406 147.289
2  subj_3  61.088  66.971
3  subj_4 169.369 175.252
4  subj_6  98.898 104.781
5  subj_7  86.782  92.665
[1] "--- Model Comparison (summed across subjects) ---"
    Model sum_AIC sum_BIC
1 ra_noRA 557.543  586.96
```

To evaluate the contribution of each parameter in the Prospect model, two reduced models were fit: ra_noLA ($\lambda = 1$ fixed) and ra_noRA ($\rho = 1$ fixed). AIC and BIC values were computed for each subject and summed across all five participants. The ra_prospect model achieved the lowest sum_AIC (536.31) and sum_BIC (580.43), followed by ra_noRA (sum_AIC = 557.50) and ra_noLA (sum_AIC = 622.40). This pattern indicates that both ρ and λ are necessary parameters for explaining choice behavior so removing either one degrades model fit. Significantly, removing λ (ra_noLA) caused a much larger increase in AIC (+86) than removing ρ (ra_noRA, +21), suggesting that loss aversion plays a more critical role than risk preference in describing decisions in this task. Subj_3 showed the most striking difference: ra_prospect AIC = 42.93 vs. ra_noLA AIC = 90.55, consistent with this subject's strong loss aversion (λ = 2.47 in the ra_prospect model). Based on both AIC and BIC, ra_prospect is the best model among the three.